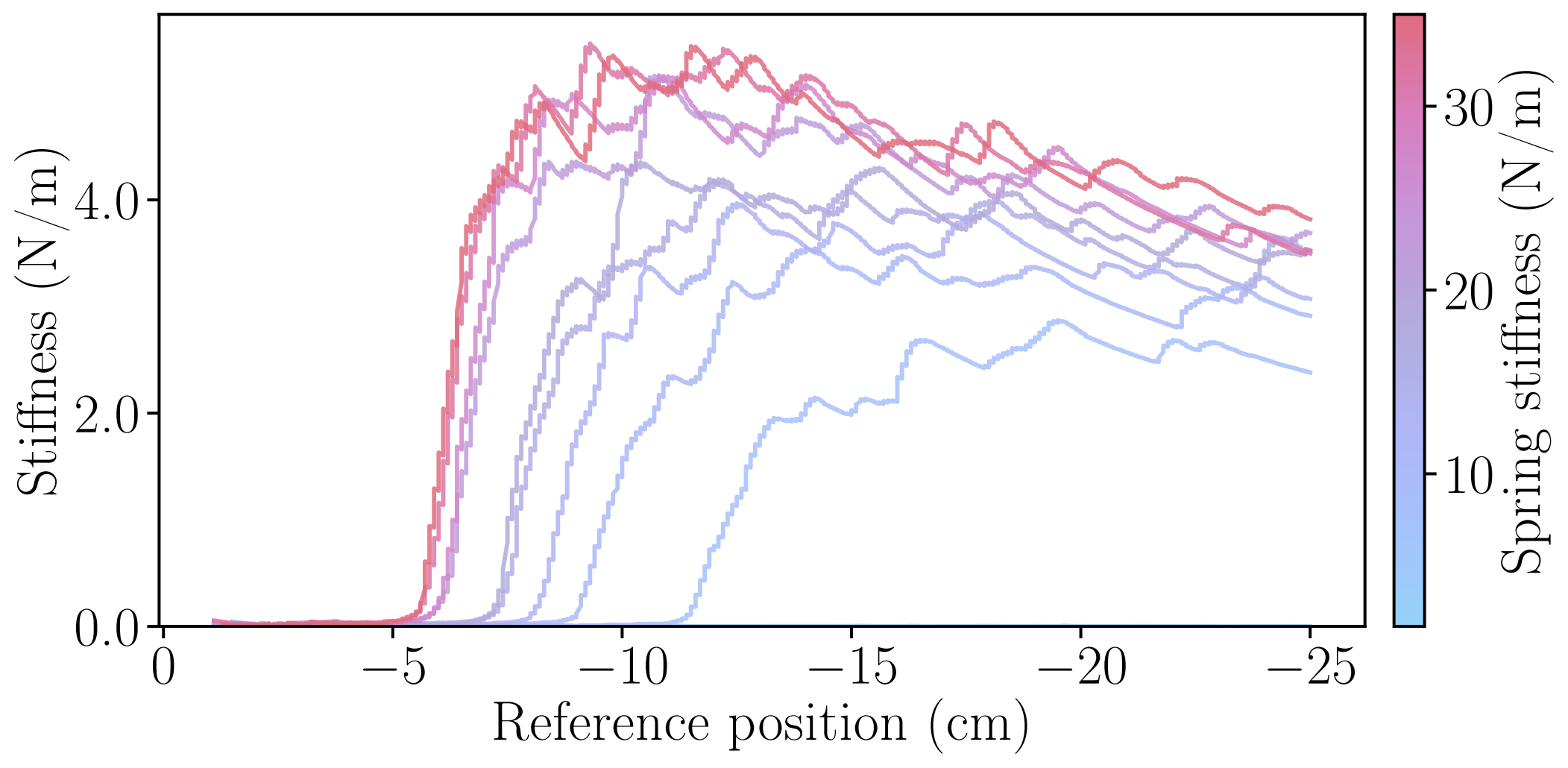

Stiffness (N/m)
4.0
2.0
0.0
0
−5
−10
−15
−20
−25
Reference position (cm)
Spring stiffness (N/m)
30
20
10